



||Jai Sri Gurudev ||

Sri AdichunchanagiriShikshana Trust®

SJB INSTITUTE OF TECHNOLOGY

Accredited by NBA & NAAC with 'A+' Grade

No. 67, BGS Health & Education City, Dr. Vishnuvardhan Road Kengeri,
Bangalore - 560 060



Department of Computer Science & Engineering

Python for Data Analytics [23CSE312]

MODULE - 1

An Introduction to Data Analysis

3rd SEMESTER - B. E

Academic Year: 2025 - 2026 (Odd)

Prepared By: Shilpashree S

Introduction to Data Analysis

- Data is everywhere: Produced by automatic systems, sensors, everyday actions (bank transactions, social media).
- Data vs. Information: Data itself is not information; it becomes useful when processed.
- Data analysis: The process of extracting actionable insights from raw data.

Purpose of Data Analysis:

- Extract hidden information: Insights that aren't immediately obvious.
- Understand systems: Study mechanisms and predict system responses.
- Make predictions: Forecast future outcomes and evolutions based on data patterns.

Introduction to Data Analysis

Evolution of Data Analysis

- **Started with simple data protection.**
- **Now a formal discipline:** Development of methodologies and models.
- **Modeling:** Translate systems into mathematical or logical forms.

Importance of Models

- **Goal of modeling:** Make accurate predictions.
- **Quality of predictions:** Depends on the choice of dataset and modeling techniques.
- **Data preparation:** Data extraction, cleaning, and preparation are crucial parts of the analysis.

Data Visualization

- **Importance:** Visual representation aids in understanding data.
- **Chart types:** Various charts (bar, line, scatter, etc.) help visualize data patterns.
- **Helps uncover hidden insights:** Visualization can reveal relationships that raw data alone cannot.

Introduction to Data Analysis

Testing and Validation

- **Model testing:** Use a different dataset (not used in the model) to check predictions.
- **Error calculation:** Measure how well the model predicts actual outcomes.
- **Assess model validity:** Compare with other models to determine performance.

Deployment of Data Analysis

- **Final step:** Implement decisions based on the model's predictions.
- **Risk prediction:** Understand risks and impacts of decisions.
- **Real-world application:** Using data analysis in decision-making improves outcomes.

Conclusion

- **Data analysis:** A powerful tool to extract insights, make predictions, and support decision-making.
- **Relevance:** Applicable in various professions.
- **Next step:** Utilize these techniques to test hypotheses and understand complex systems.

Knowledge Domains of the Data Analyst

Introduction

- **Data Analysis:** Interdisciplinary field solving problems across various domains.
- **Skills Needed:** A data analyst must have proficiency in multiple disciplines such as computer science, mathematics, and domain-specific knowledge.
- **Interdisciplinary Teams:** Larger projects often require collaboration across different expertise.

Core Knowledge Domains

1. **Computer Science**
2. **Mathematics and Statistics**
3. **Machine Learning and AI**
4. **Domain-Specific Expertise**

Knowledge Domains of the Data Analyst

Computer Science

- **Why It's Essential:** Data analysis relies on computational tools for managing, processing, and visualizing data.
- **Tools & Skills:**
 - **Programming Languages:** Python, C++, Java
 - **Software:** MATLAB, IDL
 - **Data Formats:** JSON, XML, CSV, XLS
 - **SQL & Databases:** Querying and extracting data efficiently
 - **Web Scraping:** Extracting data from websites (HTML tables, charts)

Knowledge Domains of the Data Analyst

Mathematics & Statistics

- **Key to Data Processing:** Provides the foundation for data analysis methodologies.
- **Commonly used statistical techniques in data analysis are:**
 - **Bayesian Methods**
 - **Regression Analysis**
 - **Clustering Techniques**
- **Python Libraries:** Simplify the application of complex mathematical and statistical models.

Knowledge Domains of the Data Analyst

Machine Learning & Artificial Intelligence

- **Advanced Data Tools:** Automate pattern recognition, trend analysis, and insights extraction.
- **Key Concepts:**
 - Algorithms to find patterns, clusters, and trends.
 - **Importance:** Speeds up and improves the accuracy of data insights.
- **Python Libraries:** Tools for implementing machine learning techniques.

Domain-Specific Expertise

- **Importance of Field Knowledge:** Understanding the data's origin is critical for accurate analysis.
- **Example Fields:** Biology, finance, physics, social statistics.
- **Collaboration:** Work with experts when needed to understand the data's context.

Problem-Solving Approach

- **For Smaller Problems:** Analysts must be flexible, identifying new skills or knowledge needed.
- **Solution:** Learn new methods or consult domain experts to solve issues during analysis.

Understanding the Nature of the Data

Data in Data Analysis

- **Data as the core:** The key component in all processes of data analysis.
- **Purpose:** Extract insights to increase knowledge about the system under study.

When Data Becomes Information

- **Definition of data:** Measurable or categorizable events recorded in the world.
- **Transformation:** Through analysis, data help in understanding events or making predictions.
- **Data leads to informed decisions:** Processing raw data can guide future actions.

From Information to Knowledge

- **Information:** Provides details about specific events.
- **Knowledge:** Emerges when information forms rules, enabling predictions about future events.
- **Key takeaway:** Knowledge is the ultimate outcome of successful data analysis.

Understanding the Nature of the Data

Types of Data

- **Two main categories:**
 1. **Categorical**
 2. **Numerical**

Categorical Data

- **Categorical Data:** Values or observations divided into groups.
- **Two types:**
 1. **Nominal:** No intrinsic order (e.g., colors, types of cars).
 2. **Ordinal:** Has a specific, predetermined order (e.g., rankings, education levels).

Numerical Data

- **Numerical Data:** Derived from measurements.
- **Two types:**
 1. **Discrete:** Countable, distinct, separated values (e.g., number of students).
 2. **Continuous:** Can assume any value within a range (e.g., height, temperature).

Understanding the Nature of the Data

Summary

- **Data:** The raw material of data analysis.
- **Transformation process:** Data → Information → Knowledge.
- **Understanding data types:** Helps guide analysis and improve predictions.

The Data Analysis Process

- **Data analysis** is a multi-step process for transforming raw data into insights and produce data visualizations, build predictive models, and derive actionable results.
- Data analysis is a series of interconnected stages where each plays a key role. Problem definition to deployment, each step builds on the previous one.
- Data analysis is schematized as a process chain consisting of the following sequence of stages:
 - a. Problem definition
 - b. Data extraction
 - c. Data cleaning
 - d. Data transformation
 - e. Data exploration
 - f. Predictive modeling
 - g. Model validation/test
 - h. Visualization and interpretation of results
 - i. Deployment of the solution

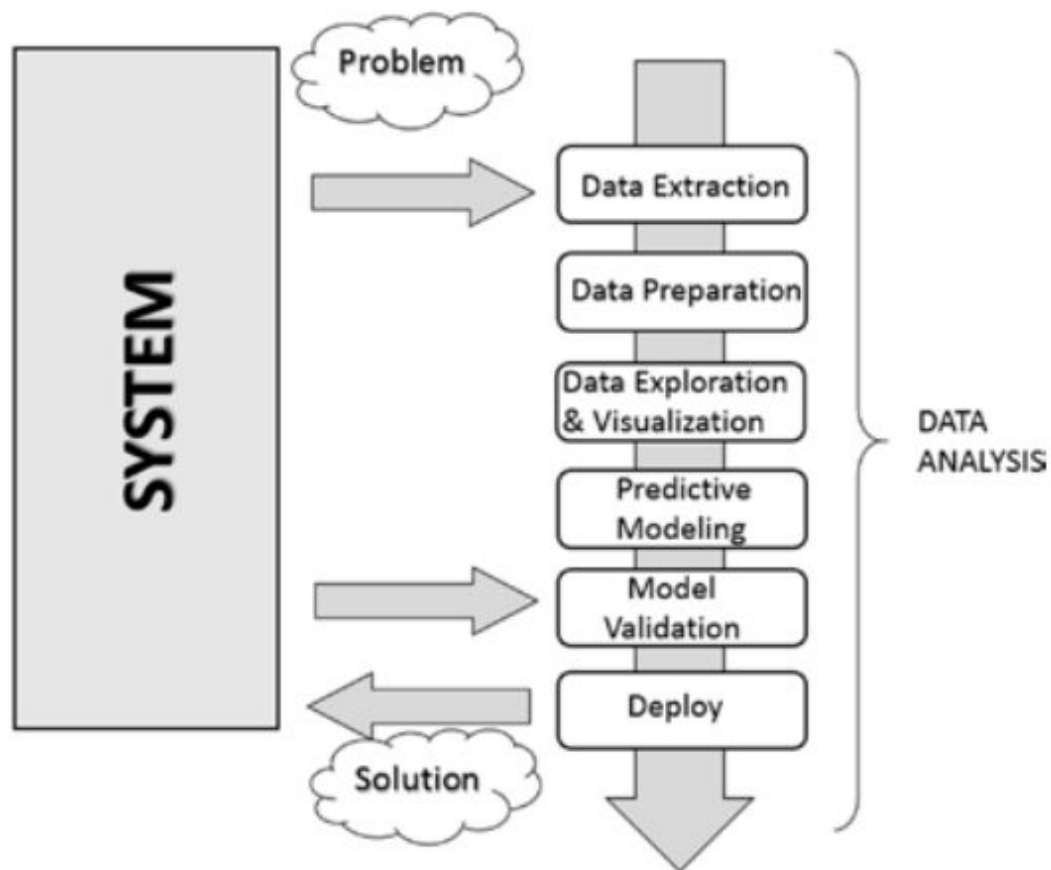


Figure 1-1. The data analysis process

Problem Definition in Data Analysis

- Data analysis begins with defining a **problem to solve**.
- The problem should relate to a **specific system, mechanism, or process** that requires understanding or optimization.
- Focus on the system's behavior to either:
 - Make **predictions** about its future behavior.
 - Make **informed decisions** to improve its functioning
- Properly document the **scientific or business problem** to:
 - Provide clarity and focus for the analysis.
 - Ensure the analysis aligns with desired outcomes.
- Once the problem is defined:
 - Begin **project planning** to identify needed resources.
 - Determine the **professionals and tools** required.
- Build a **cross-disciplinary team** for different perspectives.
- A good team is key to solving complex problems effectively.

Data Extraction

- Importance of Data:
 - **Data selection** is crucial for building a **predictive model**.
 - The data must **reflect real-world behavior** to ensure accurate analysis.
- Challenges in Data Collection:
 - Poorly chosen data can result in inaccurate models.
 - Unbalanced or unrepresentative datasets lead to poor predictions.
- Source of Data:
 - **Laboratory Data**: Experimental data are easier to identify.
 - **Real-World Data**: Can involve external experiments, **surveys**, or **interviews**.
 - **Multiple data sources** may be needed to create a comprehensive dataset.
- Data Search Techniques:
 - Web Scraping: Extracts data from HTML pages.
 - Specialized tools and software are used to gather unstructured web data.
- **Goal**:
 - To collect data that are **reliable** and **representative** for accurate predictions.

Data Preparation

1. Time and Resource Intensive:

- Data preparation is one of the most **resource- and time-consuming** steps in data analysis.
- Requires integrating data from **multiple sources**, each with **different formats**.

2. Key Activities in Data Preparation:

- **Data Cleaning**: Removing **invalid, ambiguous, or missing values**.
- **Normalization**: Ensuring data are in a consistent **format**.
- **Transformation**: Converting data into an **optimized, tabular format**.

3. Challenges in Data Preparation:

- Handling **replicated fields**.
- Dealing with **out-of-range** or erroneous data.

4. Goal:

- Prepare a **clean, structured dataset** that is suitable for the **scheduled analysis methods**.

Data Exploration/Visualization

Purpose of Data Exploration:

- A preliminary examination to **understand patterns, relationships, and trends** in the data.
- Guides in determining the most suitable **analysis methods** for model building.

Generally, this phase, in addition to a detailed study of charts through the visualization data, may consist of one or more of the following activities:

- Summarizing data Grouping data: Reducing complexity without losing key insights.
- Exploration of the relationship between the various attributes: Finding common attributes and **organizing data** into meaningful groups.
- Identification of patterns and trends: Identifying **trends, correlations, and anomalies**.
- Construction of regression models & Construction of classification models: Constructing models to predict and categorize.

Importance of Data Visualization:

- **Transforms raw data** into easily understandable **charts, graphs, and visual forms**.
- Highlights patterns and **relationships** not easily seen in raw data.
- Tools like **decision trees** and **association rules** further enhance data interpretation.

Predictive Modeling

Predictive modeling is a data analysis process used to create or select a **statistical model** that predicts the **probability** of future outcomes.

Main Objectives:

- **Prediction:** Using models to forecast future data values (**Regression Models**).
- **Classification:** Categorizing new data into pre-defined groups (**Classification Models**).
- **Descriptive:** Grouping data by shared characteristics (**Clustering Models**).

Types of Predictive Models:

- **Classification Models:** Results in **categorical** outcomes.
- **Regression Models:** Results in **numeric** predictions.
- **Clustering Models:** Provides descriptive groupings of data.

Predictive Modeling

Common Modeling Techniques:

- **Linear Regression:** Predicts continuous numeric outcomes.
- **Logistic Regression:** Predicts categorical outcomes.
- **Decision Trees:** Classifies or predicts based on feature splits.
- **k-Nearest Neighbors (k-NN):** Classifies or predicts based on proximity to known data points.

Model Selection:

- a. Different models are suited for different data types and goals.
- b. Some models provide transparent insights (e.g., **linear regression**), while others may act as a "black box" (e.g., **deep learning**).

Model Validation

- Model validation is the process of **testing** the predictive model against **new data** to assess its accuracy and generalization to unseen situations.

Training vs. Validation Sets:

- **Training Set:** Data used to **build** the model.
- **Validation Set:** Data used to **test** and **validate** the model's accuracy on new or unseen data.

Evaluation Through Comparison:

- Compare model predictions with **actual system data** to assess errors and limitations.
- Helps determine **validity range**: The model may perform well only within certain value ranges.

Model Validation

Key Techniques for Validation:

- **Cross-Validation:**
 - The training set is split into **multiple parts**.
 - Each part is used as a validation set while others are used for training.
 - Iterative process ensures **refinement** and minimizes overfitting.
- **Outcome of Validation:**
 - Quantitatively evaluates model **effectiveness**.
 - Enables **comparison** with other models to select the most accurate.

Deployment

Deployment is the **final phase** of the data analysis process, where the results are **put into practice** to provide value.

Business and Technical Outcomes:

- In **business**, deployment delivers **actionable insights** for decision-making.
- In **technical/scientific** contexts, deployment results in **design solutions** or **publications**.

Types of Deployment:

- **Report Creation:** Data analysts provide a report summarizing:
 - **Analysis Results**
 - **Decision Recommendations**
 - **Risk Assessments**
 - **Business Impact Measurement**

Deployment

Predictive Models Deployment:

- Predictive models can be deployed as:
 - **Standalone applications**
 - Integrated into **existing systems** for automation or optimization.
- **Key Focus:**
 - Translating data insights into real-world **benefits** through client or management **decisions** and solutions.

Quantitative and Qualitative Data Analysis

Quantitative Data Analysis:

- Focuses on **numerical** or **categorical** data.
- Involves **structured data** with logical order and categories.
- **Mathematical models** and **statistics** are used to derive **objective conclusions**.
- Commonly applied in **scientific**, **financial**, and **technical** analyses.

Qualitative Data Analysis:

- Deals with **unstructured** data (e.g., text, images, audio).
- Often relies on **ad hoc methodologies** to extract meaning.
- Conclusions can involve **subjective interpretation**.
- Frequently used in studying **social phenomena** or **complex systems**.

Quantitative and Qualitative Data Analysis

Key Differences:

- **Quantitative Analysis:**
 - **Objective** and data-driven.
 - Results in **quantitative predictions** (e.g., regression models).
- **Qualitative Analysis:**
 - Often **subjective** and exploratory.
 - Aims to understand **complex systems** with descriptive insights.

Applications:

- **Quantitative:** Measuring business performance, forecasting, engineering studies.
- **Qualitative:** Social research, user experience, content analysis.

Figure 1-2 shows the differences between the two types of analysis:

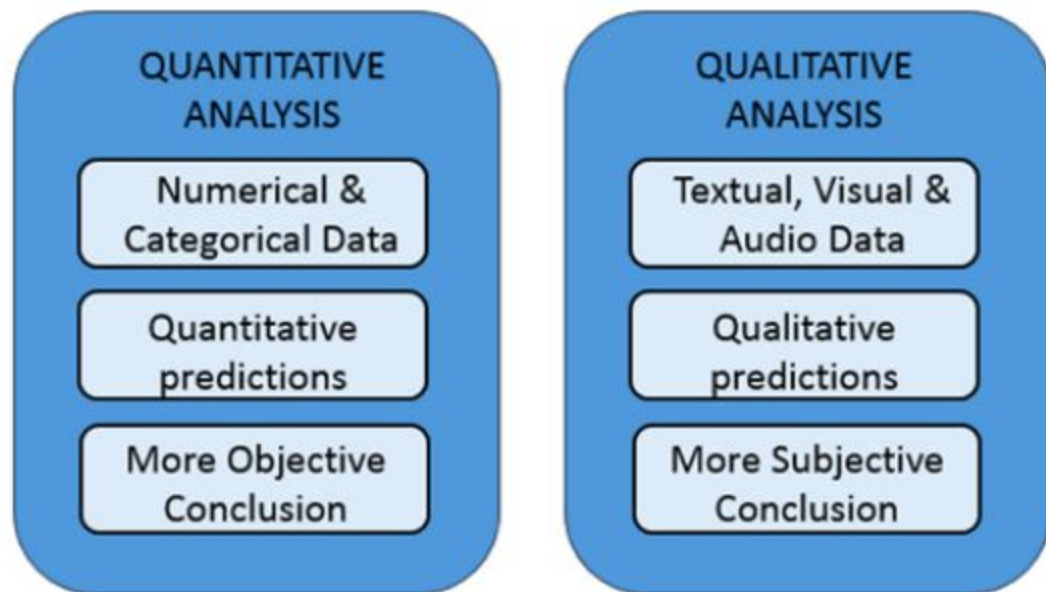


Figure 1-2. *Quantitative and qualitative analyses*

Open Data

Here is a list of some Open Data available online. You can find a more complete list and details of the Open Data available online in Appendix B.

- DataHub (<http://datahub.io/dataset>)
- World Health Organization (<http://www.who.int/research/en/>)
- Data.gov (<http://data.gov>)
- European Union Open Data Portal (<http://open-data.europa.eu/en/data/>)
- Amazon Web Service public datasets (<http://aws.amazon.com/datasets>)
- Facebook Graph (<http://developers.facebook.com/docs/graph-api>)
- Healthdata.gov (<http://www.healthdata.gov>)
- Google Trends (<http://www.google.com/trends/explore>)
- Google Finance (<https://www.google.com/finance>)
- Google Books Ngrams (<http://storage.googleapis.com/books/ngrams/books/datasetv2.html>)
- Machine Learning Repository (<http://archive.ics.uci.edu/ml/>)

Python and Data Analysis

The focus here is on using **Python** to develop all data analysis concepts. Python has become a popular programming language in scientific and data circles because it offers a wide array of tools for analysis and data manipulation.

Why Python Over Other Languages?:

- While languages like **R** and **Matlab** are also used for data analysis, Python stands out because it's not just a tool for data processing but offers **unique advantages**:
 - Python has a growing ecosystem of **libraries** that make advanced data analysis easier and more efficient. Examples include NumPy, pandas, and Matplotlib.
 - It can **interface with other languages** like **C** and **Fortran**, meaning it can leverage even more power for specific tasks.

.

Python and Data Analysis

More than Just Data Analysis:

- Unlike specialized languages that are solely used for data (like R), Python is versatile. You can use it for **general programming**, creating **scripts**, interacting with **databases**, and even **web development** (via frameworks like Django).
 - Example: You can build a data analysis project and integrate it into a **web application**—something harder to do with languages like R

A Future-Proof Language:

- Given its flexibility, expanding libraries, and powerful tools, **Python** is considered a smart choice for anyone looking to dive into **data analysis**.
 - It's not just a current trend—it's likely to remain an essential tool for data analysts in the future.

Reference:

Text Book 1 Chapter 1: Python for Data Analytics, Fabio Nelli.